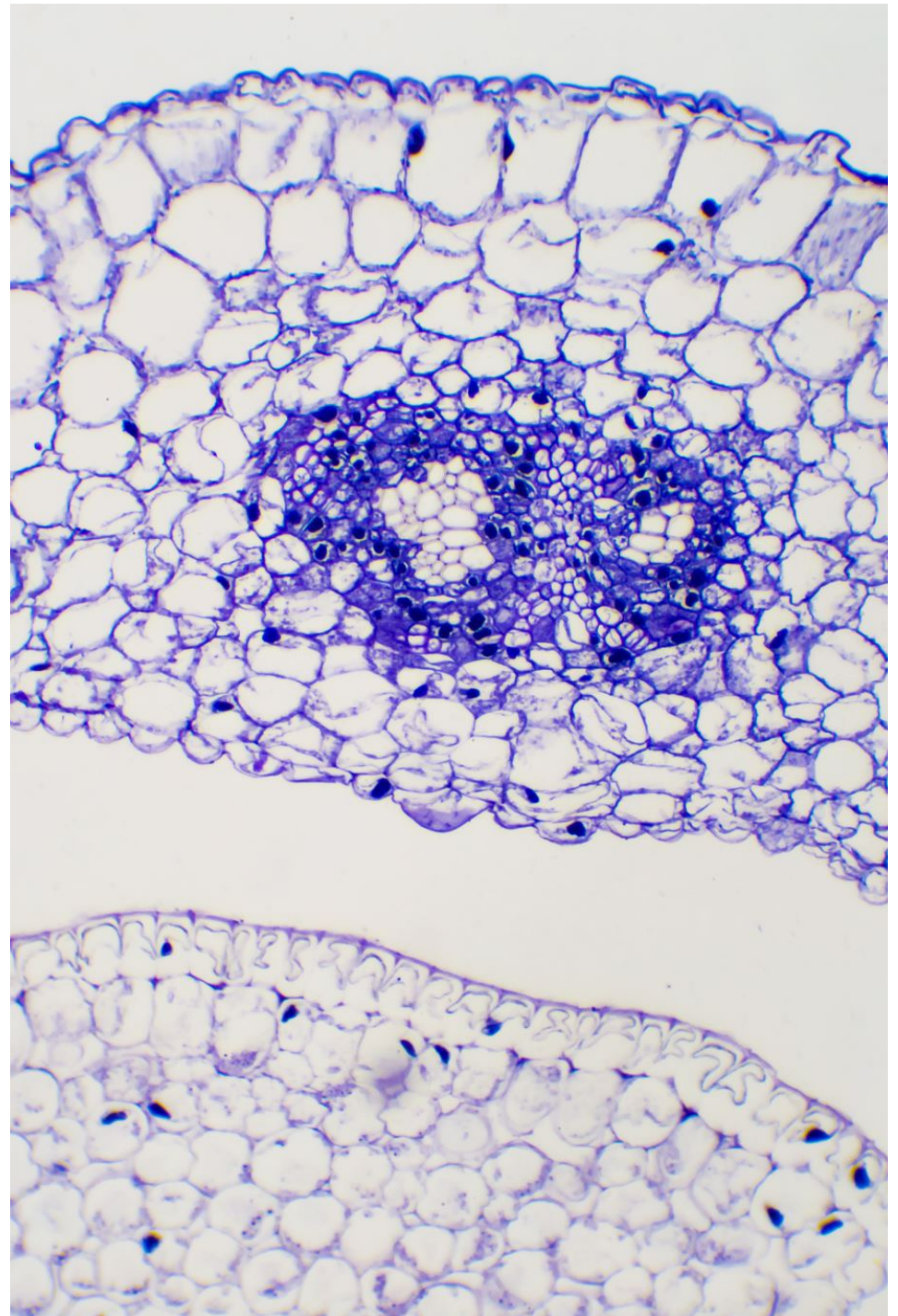


Capstone 3: Teledermatology Skin Lesion Classification

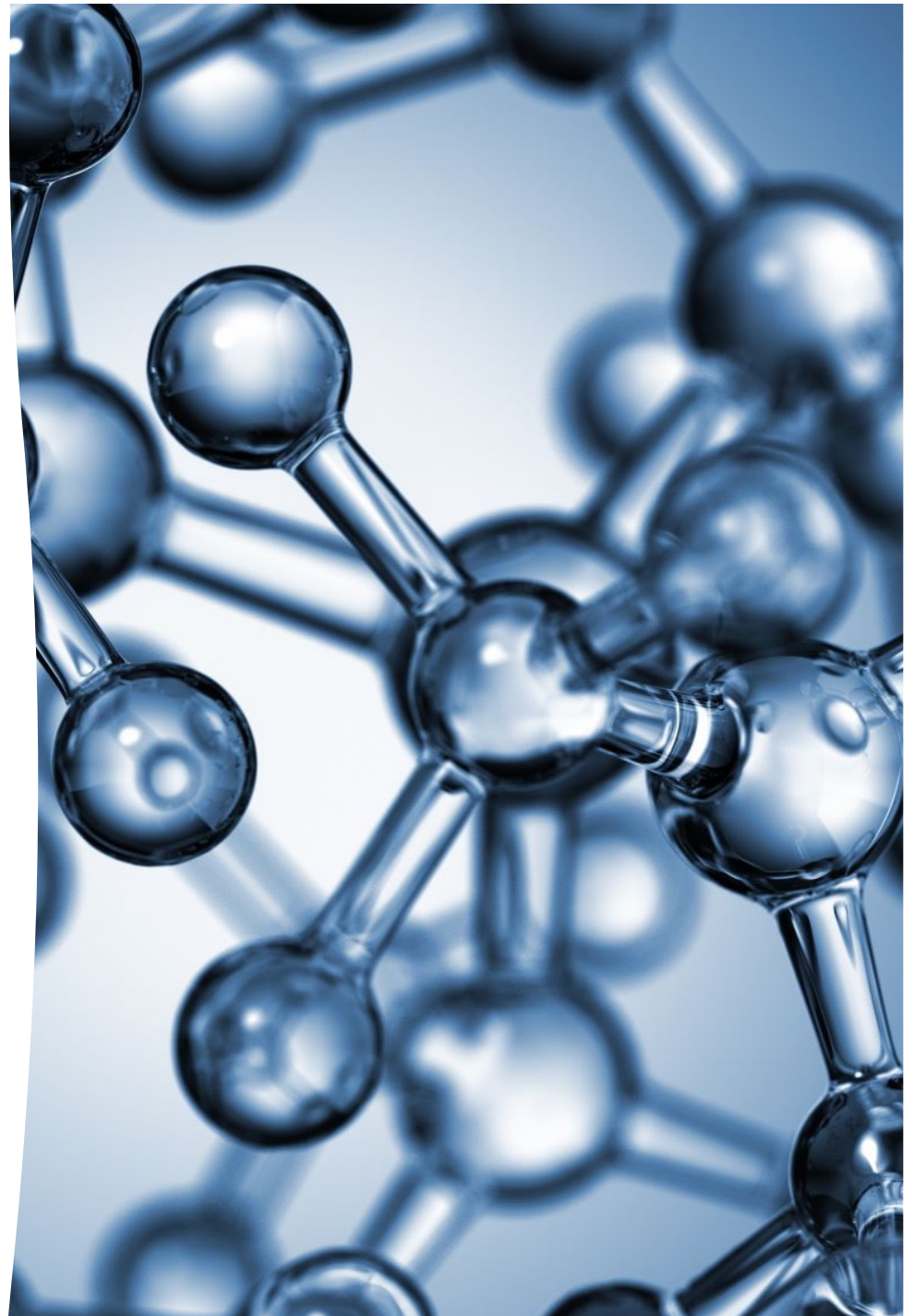
Manuel Ramirez
Chimarro



Problem Statement & Context

Dermatological triage, especially in remote areas, can benefit from AI to classify skin lesions.

Goal: Support teledermatology diagnosis using machine learning on the HAM10000 dataset.



Stakeholders & Success Metrics



Stakeholders: Dermatologists, Clinics,
Remote Healthcare Systems.

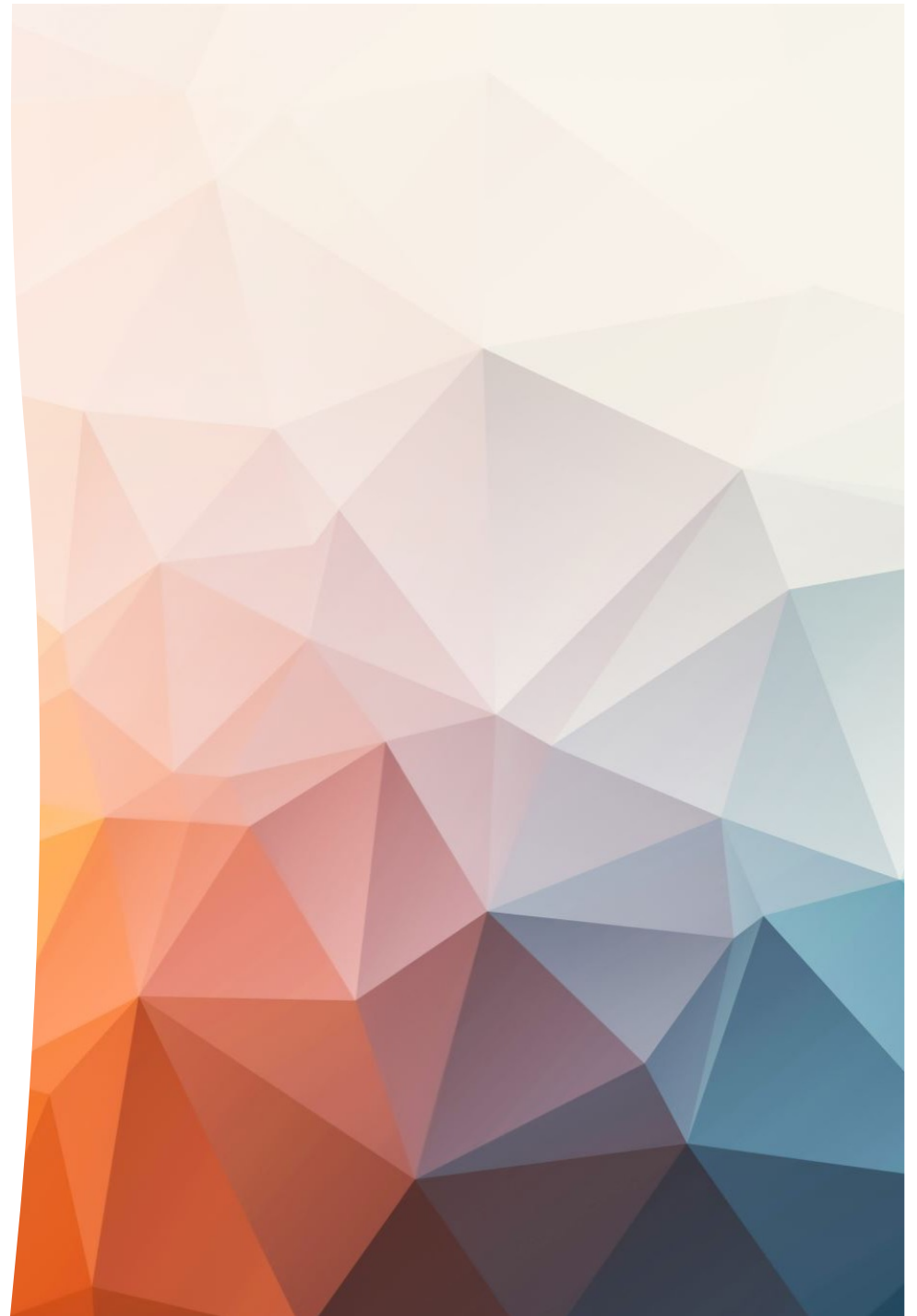


Success: Improve classification accuracy
on rare lesion types (melanoma, vascular)
with >75% overall accuracy.

Dataset Overview

HAM10000: ~10,000 dermoscopic images labeled with 7 skin lesion types.

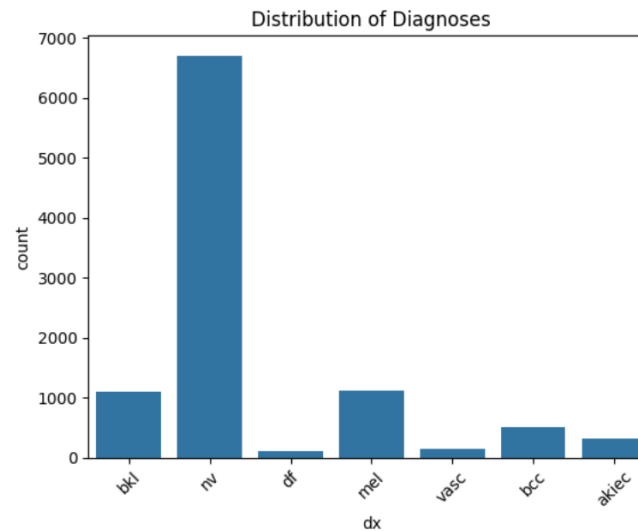
Metadata includes patient age, sex, lesion location, and diagnosis type.



- Dataset Overview”

- The dataset is heavily imbalanced — with over 67% of samples being nv (melanocytic nevi), while critical classes like mel (melanoma) and vasc (vascular lesions) are underrepresented. This imbalance necessitated the use of **focal loss** and **data augmentation** in modeling.

```
# Display numeric class distribution
print(metadata['dx'].value_counts())
print(metadata['dx'].value_counts(normalize=True) * 100)
```



```
dx
nv      6705
mel     1113
bkl     1099
bcc      514
akiec   327
vasc    142
df       115
Name: count, dtype: int64
dx
nv      66.949576
mel     11.113330
bkl     10.973540
bcc      5.132302
akiec    3.265102
vasc     1.417873
df        1.148278
Name: proportion, dtype: float64
```

Data Wrangling

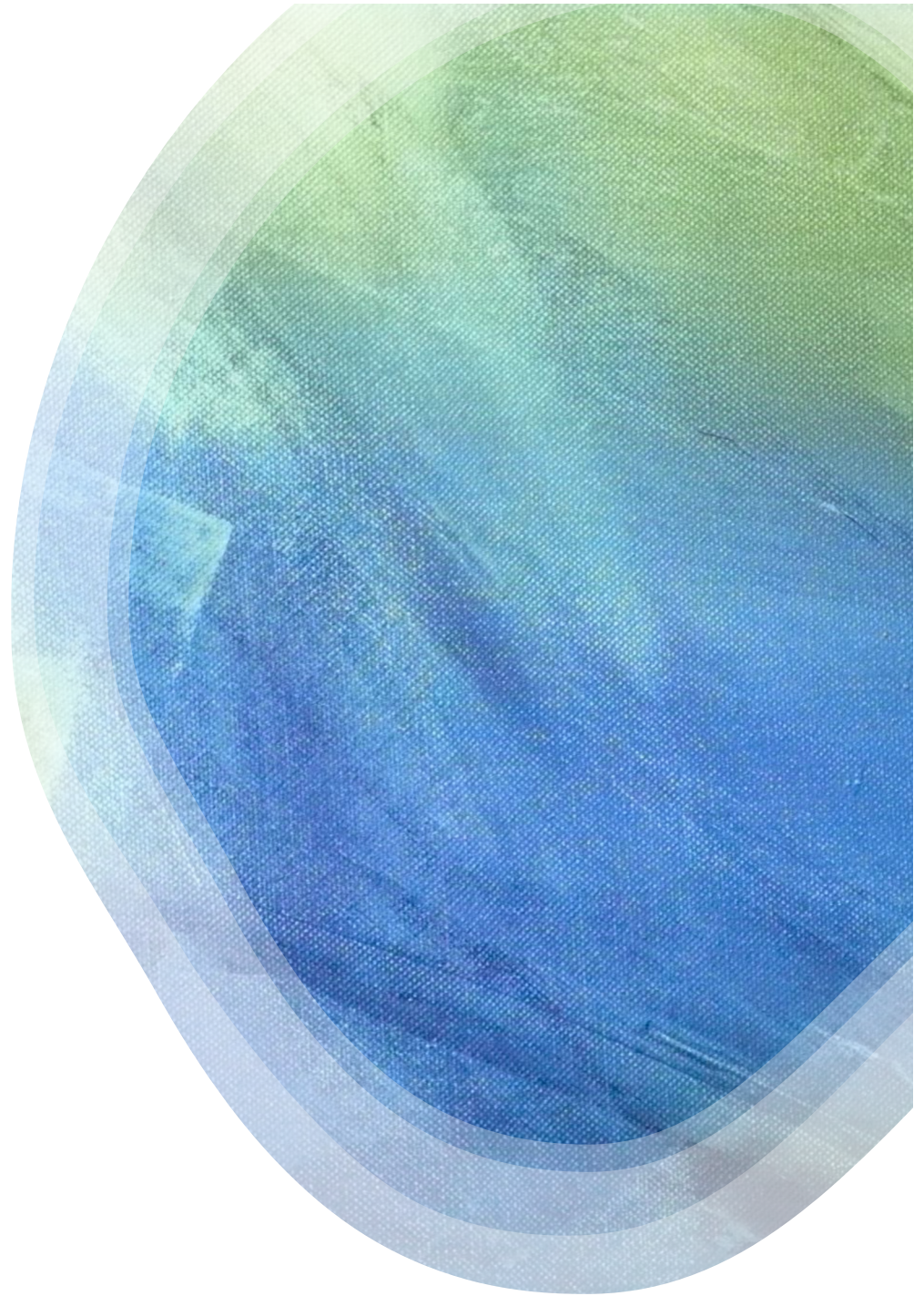
- Merged image metadata
with filenames



- Encoded categorical
variables (sex, localization)



- Handled missing values
and standardized features

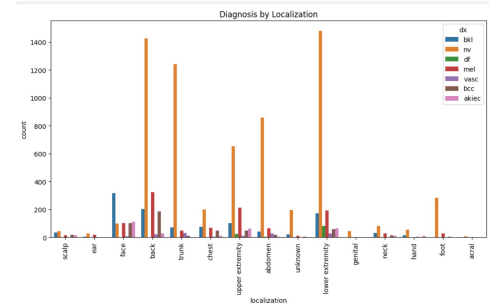
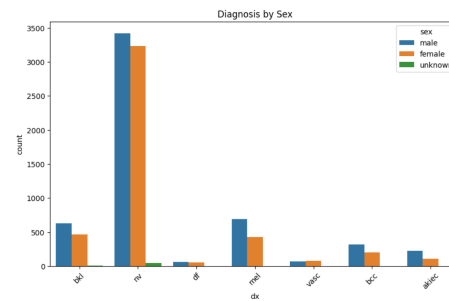
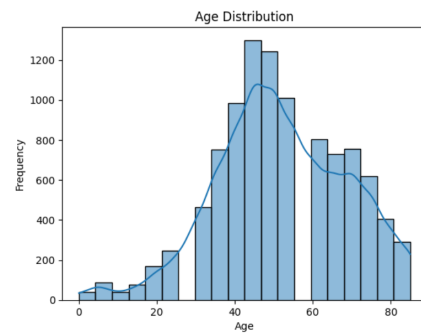


Exploratory Data Analysis

- - Imbalanced class distribution (e.g., 'nv' overrepresented)
- - Age distribution skewed older
- - Some lesion types had fewer than 100 examples



Exploratory Data Analysis



Modeling Approach



- Baseline: Logistic Regression, Random Forest on metadata



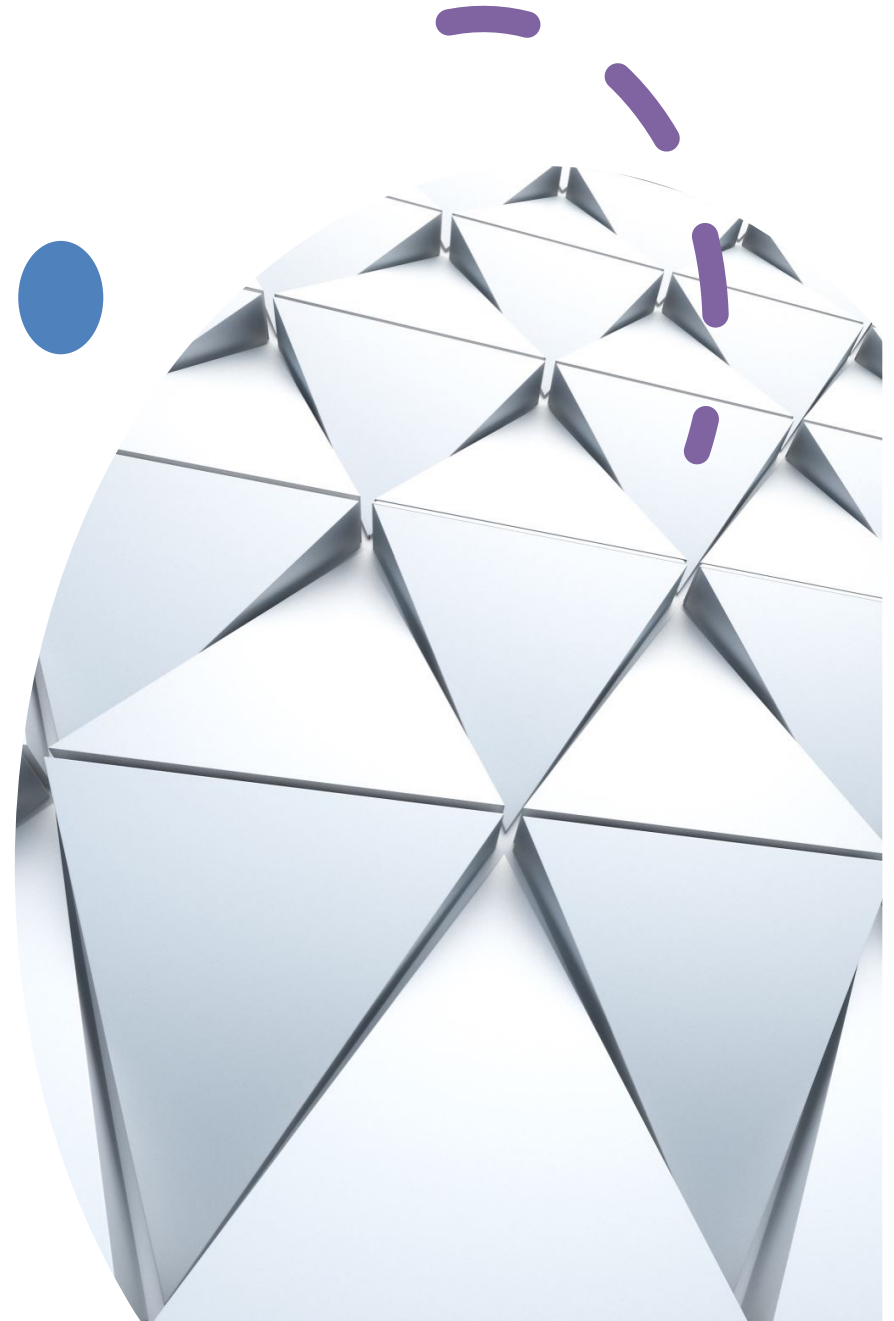
- CNN on images



- EfficientNetB3 pretrained model fine-tuned



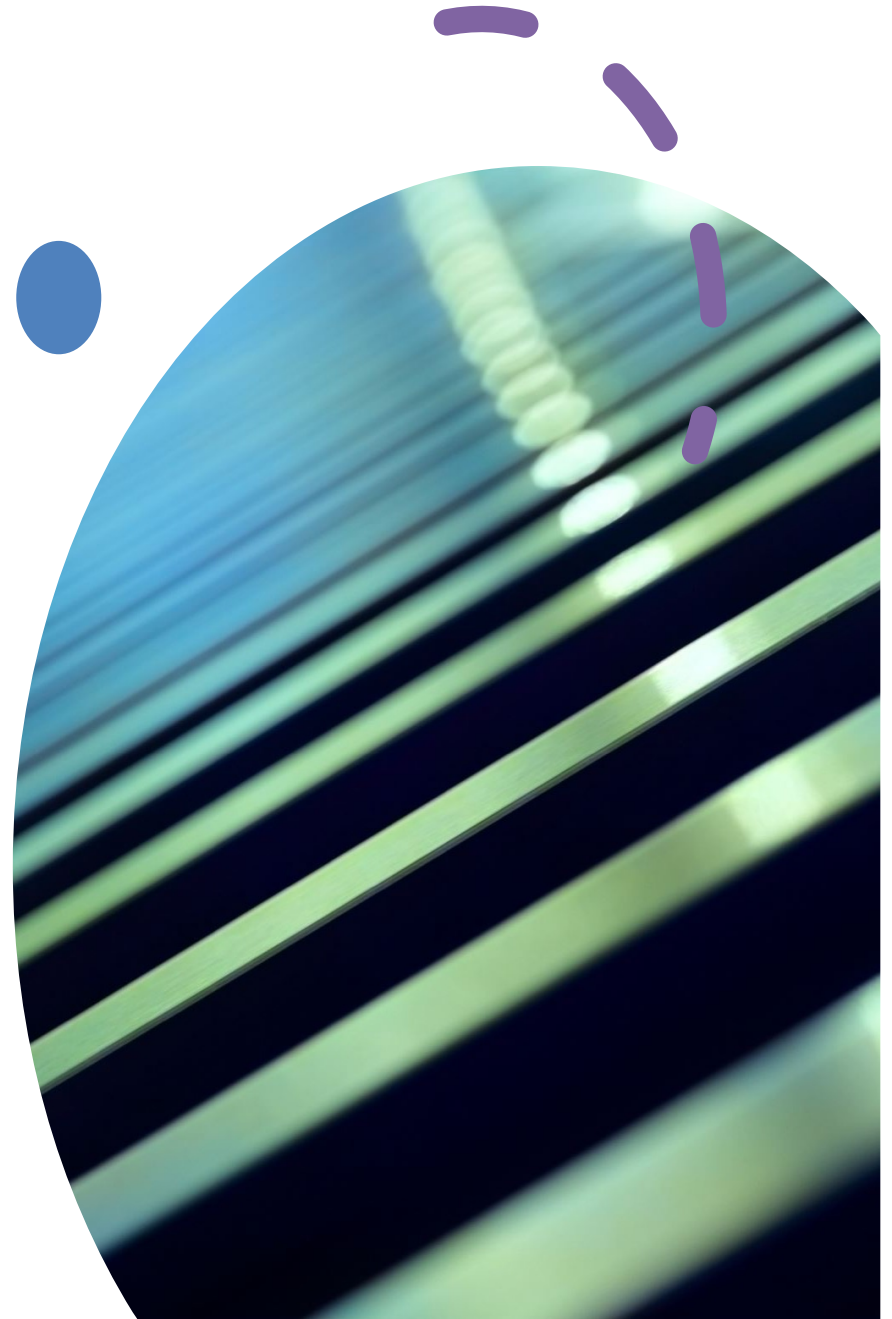
- Final Model: Multimodal CNN + Metadata



Loss Function & Augmentation

- Focal Loss ($\alpha=1.0$, $\gamma=2.0$) used to handle class imbalance

- Strong data augmentation: rotation, flip

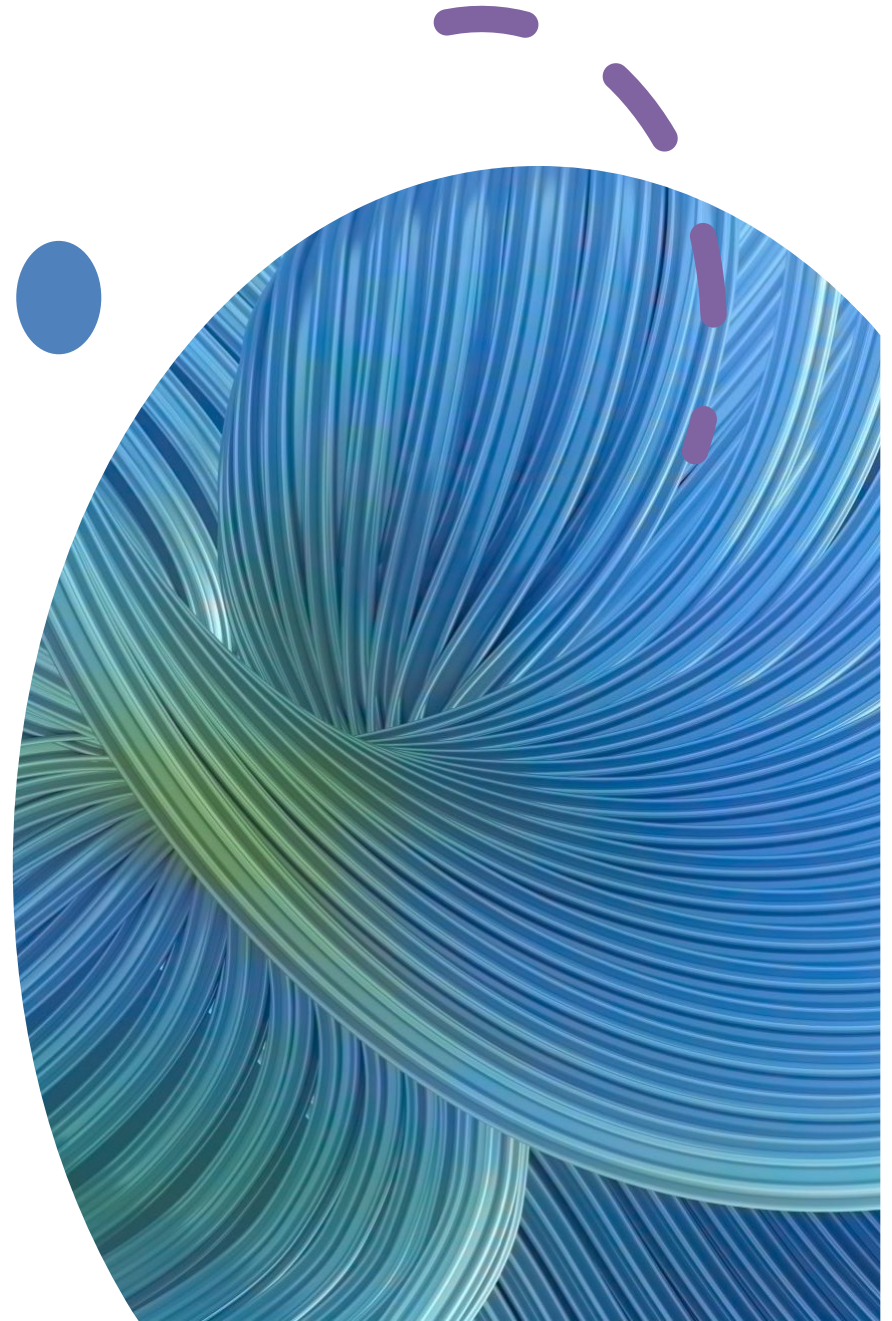


Final Model Performance

EfficientNetB3 outperforms (64%) the previous efficientNetB0 model of 46% of accuracy.

- Improved recall for minority classes

- Multimodal input outperformed image-only models



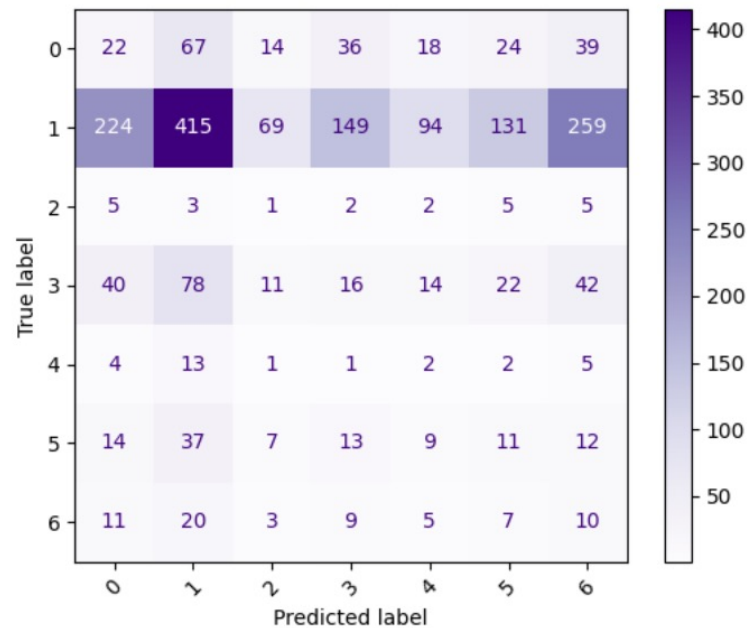
- EfficientNetB0 model Accuracy

00/00 000 0.00ms/step

Classification Report (After Fine-Tuning):

	precision	recall	f1-score	support
0	0.07	0.10	0.08	220
1	0.66	0.31	0.42	1341
2	0.01	0.04	0.02	23
3	0.07	0.07	0.07	223
4	0.01	0.07	0.02	28
5	0.05	0.11	0.07	103
6	0.03	0.15	0.05	65
accuracy			0.24	2003
macro avg	0.13	0.12	0.10	2003
weighted avg	0.46	0.24	0.30	2003

[46]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7f976739f2e0>



EfficientNetB3 model accuracy

63/63  17s 274ms/step

EfficientNetB3 Classification Report:

	precision	recall	f1-score	support
akiec	0.34	0.53	0.42	220
bcc	0.86	0.79	0.82	1341
bkl	0.00	0.00	0.00	23
df	0.22	0.34	0.27	223
mel	0.05	0.04	0.04	28
nv	0.15	0.03	0.05	103
vasc	0.17	0.12	0.14	65
accuracy			0.63	2003
macro avg	0.26	0.26	0.25	2003
weighted avg	0.65	0.63	0.64	2003



Visual Results

- Example predictions visualized for interpretability.
- Grid of true vs predicted labels helps evaluate model insights.

Model loaded successfully.
1/1 ————— 3s 3s/step
1/1 ————— 0s 82ms/step
1/1 ————— 0s 79ms/step
1/1 ————— 0s 108ms/step
1/1 ————— 0s 68ms/step
1/1 ————— 0s 62ms/step
1/1 ————— 0s 100ms/step
1/1 ————— 0s 86ms/step
1/1 ————— 0s 80ms/step

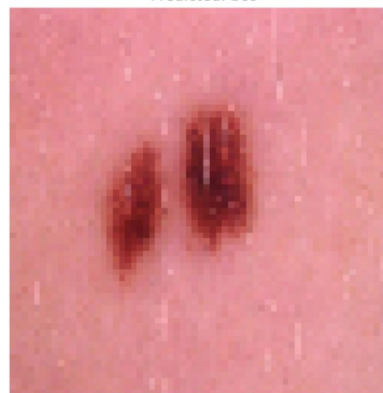
Predicted: bcc



Predicted: vasc



Predicted: bcc



Predicted: df



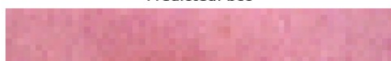
Predicted: akiec



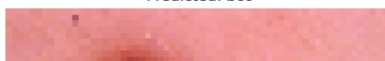
Predicted: akiec



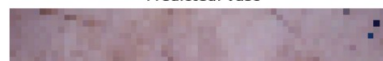
Predicted: bcc



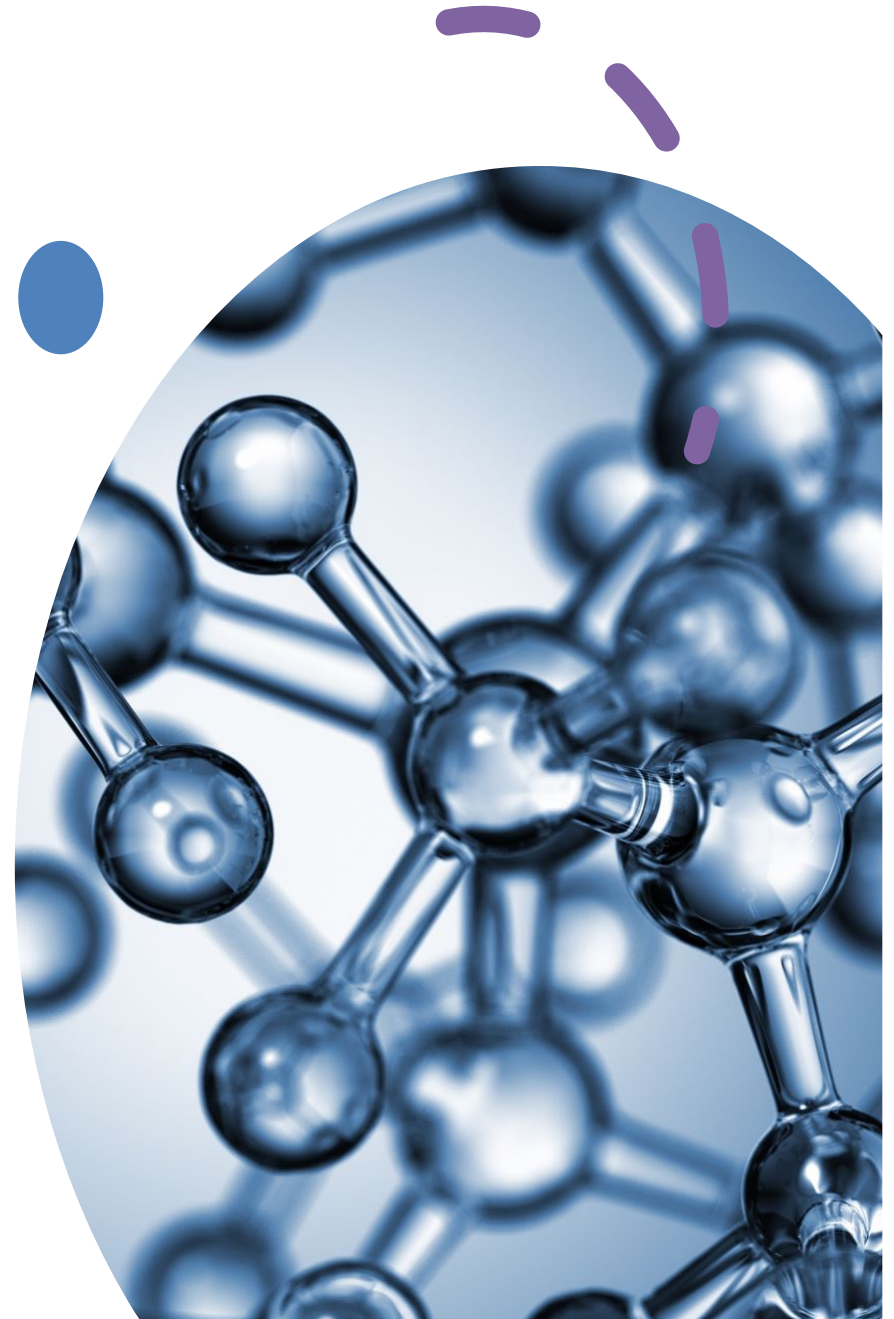
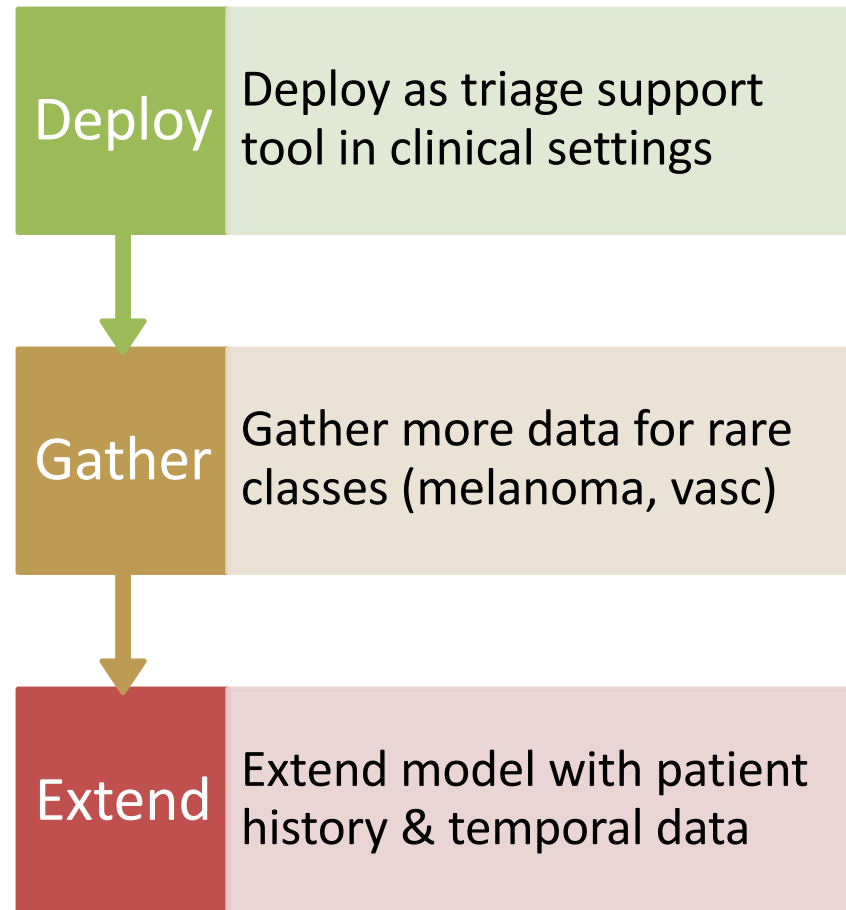
Predicted: bcc



Predicted: vasc



Recommendations



Future Work

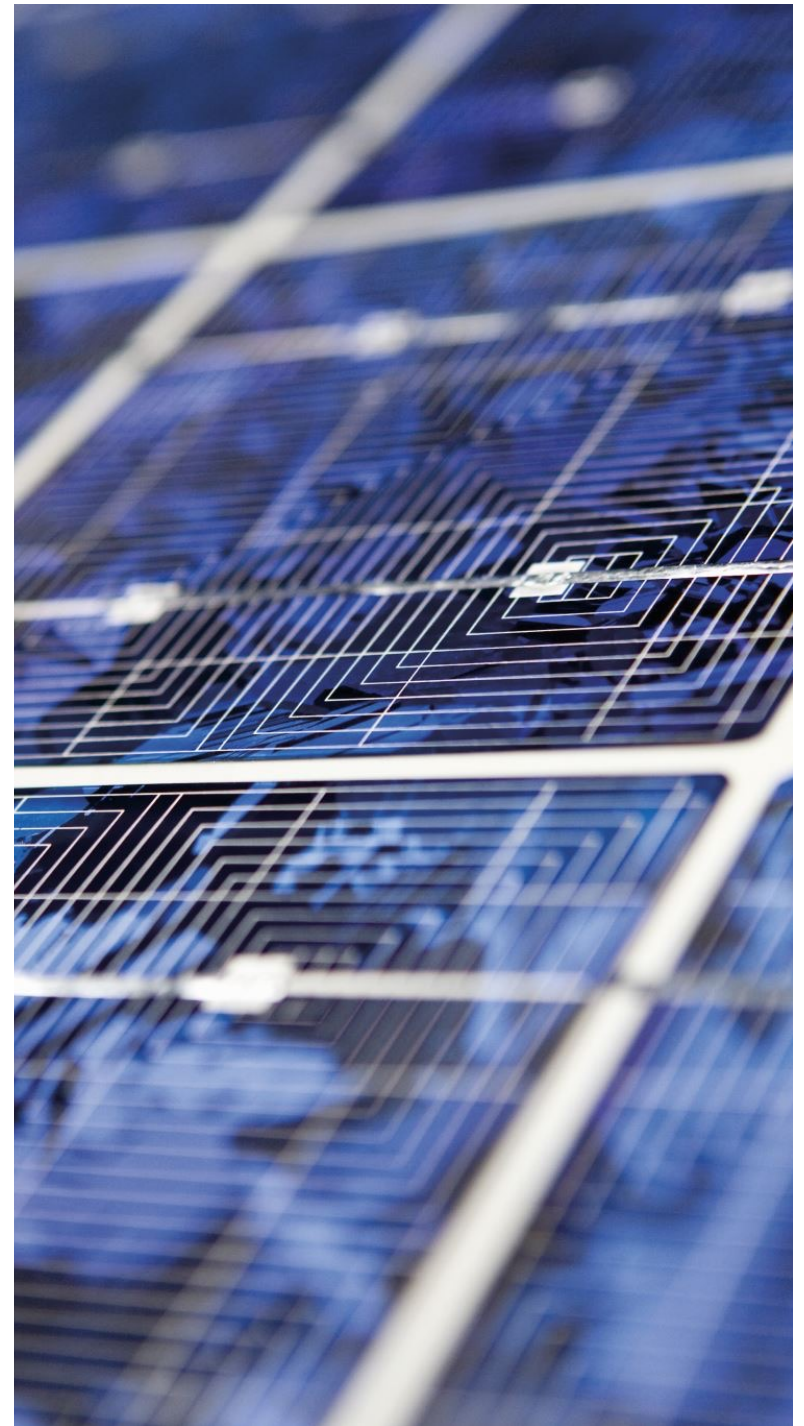
Tune focal loss hyperparameters per class

Apply skin tone normalization or lesion boundary segmentation

Metadata improvements with richer features

Advanced Data Augmentation

Multimodal pipeline enhancements



Thank You!



Questions?



Presentation prepared
for Springboard
Capstone 3



Mentor Session: Final
Review

